7	**(a)**			
		photon (of energy equal to the energy difference) is emitted when <u>electron de-excites from higher to lower energy level</u> B1 <u>each line corresponds to specific wavelength</u> B1 energy of photon <u>depends on frequency or wavelength only</u>, such that E=hf, so <u>energy change is discrete</u> B1 these imply discrete electron energy levels exist in atoms		
	(b)			
		(i)	energy (depends on frequency) <u>equal</u> for each photon B1 <u>maximum</u> kinetic energy for electron emitted from surface = energy of photon – work function energy B1 <u>lower energies (than maximum)</u> <u>when energy required to bring electron to surface</u> B1	
		(ii)	**1.** $hf_{\text{min}} = \frac{hc}{\lambda_{\text{max}}} = 3.2e$ maximum wavelength $= \frac{6.63\times10^{-34}\times3\times10^{8}}{3.2\times1.6\times10^{-19}} = 388$ nm C1 wavelengths causing emission are <u>297 nm</u> and <u>365 nm</u> A1	
			2. intensity <u>proportional/affect rate of photons arriving/incident on surface</u> B1 photon <u>energy remains the same</u> B1 so, no effect	

8	**(a)**	**(i)**	mean/average squared speed of the gas molecules	B1	